

OPTT Analysis

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Abstract

Founded in 1894, Ocean Power Technologies (OPTT) is a maritime-focused company with an emphasis on finding solutions to the need for more efficient/productive infrastructure. Their services span defense and security, oil and gas procurement, and off-shore wind markets.

OPTT went public on the NASDAQ on the 25th of April, 2007. Initially, the stock exploded, with a closing price high of \$3,652.00 on November 12th, 2007. Ever since, the stock has been on the decline, falling under \$5 per share in early 2019. Crossing that \$5 threshold means the stock is now classified as a penny stock. Ocean Power Technologies hosts a market cap of \$74.6 Million (at the time of writing), thus designating the company as a micro-cap company.

Since approximately 2012, OPTT has suffered in many ways that have led to investor distrust. The three main areas of concern are the high risk designation due to high volatility, share dilution, and multi-million dollar operating losses. For example, between 2012 and 2026, the number of shares (approximately) increased from 5 million to 180 million while total operating losses during that time exceeded \$300 Million.

That being said, OPTT is not entirely out of the market. On January 6th, 2026 OPTT announced they had been awarded a contract by the U.S. Department of Homeland Security for the purpose of supplying the U.S. Coast Guard with their new PowerBuoy systems. The contract was valued at \$6.5 Million. This contract was not only a lifeline for OPTT but marked the transition from a sustainability focused private company into what is now essentially a revenue focused defense contractor.

In this project, I use statistical modelling and financial metrics to analyse OPTT's performance and risk overtime. From there I aim to make inferences about whether thus far the contract has begun to aid OPTT recover. I then test my inferences using machine learning algorithms to predict future stock prices.

I will be using OLS regression models for performance analysis, using the log of Volume as the dependent variable. I am using volume because in contrast to the other metrics I can choose from, Volume is able to differentiate between direction and attention in the market. Volume focuses on how many shares are bought or sold, meaning it is a good measure of general activity

surrounding the stock itself as opposed to what the public think of the stock. I take the log of Volume due to it being right-skewed with a median of 7,710 vs a mean of 1,974,392.

I will use a logistic regression model for risk analysis, using a “risk event” binary as the dependent variable. The risk event will be triggered when a significant price drop or sell-side volume spike occurs, or both. The models will be ordinal in nature, each with 3 “severity” tiers, defined as:

- **Level 1** -> No Risk Event: Price Drop < 3% and Volume Multiple < 1.25
- **Level 2** -> Moderate Risk Event: Price Drop between 3% - 7% (inclusive) or Volume Multiple between 1.25 - 2 (inclusive) AND neither metric surpasses Level 3 thresholds.
- **Level 3** -> High Risk Event: Price Drop > 7% or Volume Multiple > 2

I will then move into predictive modelling, using Random Forest, XGBoost, and LSTM Networks to predict future performance.

For the LSTM Network, the adjusted closing price will be the outcome variable. I am using adjusted closing over opening or closing price due to the notion that an ML algorithm will look at fluctuations between day as catastrophic economic events if the drop is significant enough from one day to another. It is also of note that if that drop is caused by a volatility spike, the adjusted closing price will catch it. Moreover, the adjusted closing price would back-trace historical data which would prevent the model from falling for false inter-day price gaps.

For the Random Forest and XGBoost algorithms, it does not make sense to predict price as a raw dollar metric due to the lack of trend-awareness that these algorithms possess. Both algorithms function by breaking the data down into rectangular cross sections based on real historic data. Thus, these models will never predict a future dollar price that exceeds the upper and lower bounds of the data - a reality that does not exist in the stock market. As a solution, I will predict returns using a calculated percent change metric as the outcome variable. This percent change will be calculated using adjusted closing price

To conclude, I will compare the effectiveness of the 3 ML models using error scores to determine which stand-alone algorithm performed best.

Why does any of this matter? Firstly, I enjoy machine learning and investing, so this was a natural project for me to take on. Secondly, it is an interesting question to ask - can traditional ML handle the noise of penny stocks? Currently, analysts rely on hybrid algorithms that incorporate classification models, and sentiment analysis. In most cases, LSTM's tend to confuse the noise of unexpected price spikes that are overly common in penny stocks for true trends, and I want to see the comparison myself.

The Data

In order to take on this project, I utilise a variety of methods to procure data. The main pathway is the `tidyquant` package from which I am able to download historic data about the desired ticker. `tidyquant` primarily leans on Yahoo Finance for key metrics, but also pulls from several other databases such as the Federal Reserve Economic Data (FRED).

I also make queries to NewsAPI and Finnhub, two APIs that allow me to scrape for media related to my chosen ticker in order to perform sentiment analysis in conjunction with the machine learning models. Unfortunately, there is limited results for Ocean Power Technologies and as such I do not continue down this path.

Using `tidyquant`, I retrieve data from April 25th, 2007 (IPO) until June 21st, 2026 which is the date of writing, although the data becomes usable after row 34 due to the staggered system in which some calculated metrics such as MACD work. That being said, the complete dataset boasts 4,819 observations across 17 variables (22 when including ticker name, year, month, month name, and day).

Of these 17 variables, some notable measures include:

- **open_price**: Stock price at open on a given trading day
- **close_price**: Stock price at close on a given trading day
- **Volume**: Number of shares traded (bought or sold) during a given trading day
- **true_range**: Total price volatility by finding the greatest distance between current high, current low and previous closing price
- **ATR**: Moving average of true range over a 14-day period
- **rsi**: Relative Strength Index: Measures magnitude and speed of price fluctuations on a scale of 0-100 (momentum indicator)
- **macd_hist**: Measures if the stock's momentum is speeding up or slowing down
- **obv**: On-Balance Volume: Relationship metric between volume and price used to spot divergences

To illustrate the structured dataset, rows 34–39 are displayed below. Because several indicators rely on rolling or exponential moving-average calculations of varying lengths (e.g., a 14-day window for ATR and RSI, and a 26-day EMA - Exponential Moving Average - for the MACD trendline, followed by a further 9-day EMA for the MACD signal line), the initial rows of the dataset default to NA values until each indicator has accumulated sufficient historical data to produce a valid output. The MACD signal line, in particular, does not return numeric values until row 34, as it requires the MACD line itself to first be calculated. Rows 34–39 are therefore displayed to ensure all indicators, including MACD and its derived histogram, are represented with complete, numeric, and non-NA values.

In short, the following table represents values from June 12th, 2007 - June 19th, 2007. June 16 and June 17 are omitted as they fall on a Saturday and Sunday which are non-trading days, meaning no data can be collected on those days.

open_price	close_price	Volume	true_range	ATR	rsi	macd_hist	obv
3036	3024	386	150	135.4286	40.01469	0.0571432	7232
3004	3128	186	132	134.0000	45.91322	0.3497787	7418
3144	3182	286	128	128.2857	48.73218	0.6559246	7704
3190	3222	236	80	126.2857	50.77863	0.9234428	7940
3240	3154	120	88	117.5714	47.32051	0.9297880	7820
3160	3284	324	246	128.0000	53.79843	1.1732797	8144

MACD is the Moving Average Convergence Divergence, essentially a tool used to determine buy/sell signals, price trends, and momentum. There are 3 main parts to MACD: the trendline, the signal line, and the histogram.

The MACD trendline is the main part of this indicator. it is calculated by subtracting a 26-period EMA from a 12-period EMA. The 26-period EMA is the longer-term trend and the 12-period EMA is the shorter-term momentum. EMA stands for Exponential Moving Average and is a metric that smooths out the price data to gauge a better view of the direction and momentum of trends. It is used over standard moving averages because it puts higher emphasis on later data points, which matter most in the stock market. Due to the volatility of the OPTT stock, I elect to use 12 and 26 period EMAs as opposed to 50 and 200 period EMAs which might be more favourable to longer term analysis or more stable stocks.

The MACD signal line is a 9-period moving average of the MACD trendline. In other words, it acts as a “slower” baseline metric for the momentum of the stock which can then be compared to the “faster” trendline.

That relationship is expressed by the MACD histogram. It is a metric that helps investors determine if it is a bullish vs bearish market, which in turn aids in determining whether or not to buy or sell the stock. I calculate this value by subtracting the macd signal from the macd trendline value for any given observation, represented as `macd_hist` in the table above.

Below are the values of the MACD histogram displayed as a histogram:

-placeholder for `macd_hist` histogram output-

Looking at the pictured histogram, we see that the data is (for the most part), normally distributed around 0 with a slight right-skew. This tells me that on average, across all time, momentum is neutral, meaning no real sustained bullish nor bearish trends. The extended tail that causes the right-skew is most likely a result of the initial bullish trend when the company first became public on the NASDAQ. This would explain why we don't see a tail of equal length on the left: the decline was slower, across a longer period of time, meaning the fluctuations during the decline will mitigate the effect of that sustained decline. On a similar stream, we would expect the MACD histogram of a higher valued, more stable stock to have a tighter distribution. For example, an S&P500 tracker like VOO could be expected to have

a tighter distribution with a slight right-shift (not skew) due to it's sustained upward trend without the extreme volatility witnessed in OPTT.

Below is the exact MACD histogram for VOO:

-placeholder for VOO MACD histogram for comparison-

Sentiment Analysis

-placeholder for data table with all 4 articles represented and brief discussion of them-

After querying both NewsAPI and Finnhub, we returned a total of 4 relevant articles. Due to concerns of limited data and subsequent biases, I will not be using sentiment analysis scores within my models.

Instead, the analysis will focus on price-derived technical indicators alongside trading volume. These variables are derived directly from empirical trading data, avoiding the sample-size and bias concerns associated with the limited sentiment data.

In order to maintain structural robustness via alternative data streams, the analysis will pivot to integrate macroeconomic corporate health metrics, specifically metrics such as outstanding share dilution and order backlog growth.

Alternate Data Procurement

-placeholder-

tq_get -> financials -> filter for type == income, unnest quarterly.

problem -> quarterly means 4 data points per year, not a lot of info (data-sparsity bias)

solution -> forward stepwise function to fill for weekly / monthly data (outstanding shares)

solution -> linear interpolation to fill for weekly / monthly data (order backlog growth)

Linear Regression Model

-placeholder for written explanation of model-

big question: does volatility relate to volume, in general, across the whole dataset

```
#regression model + output
ols <- lm(log(Volume) ~ rsi + macd_hist + shares_outstanding + order_backlog_growth, data = d)
summary(ols)

#check residuals
plot(ols)
```